

Rishav Roy

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EDUCATION

University of Wisconsin–Madison, B.S.

Triple Major: Economics, Mathematics, and Data Science

Recent GPA (since Summer 2024): 3.9 | Overall GPA: 3.3

May 2026

Madison, WI

Graduate-Level Coursework

PhD Macroeconomics I–II (MATLAB)

Top scorer in PhD Macroeconomics II

PhD Microeconomics I–II

PhD Industrial Organization I–II (audit)

PhD Econometrics I–II (Stata)

Top scorer in PhD Econometrics II in 18-cr. semester

Master's-Level Measure & Integration (Honors)

Top scorer of first half in 18-cr. semester

Select Undergraduate Coursework

Data Science Programming I–II (Python, SQL)

Introduction to Optimization (Julia)

Machine Learning & Classification (Python)

Econometrics: AI & Machine Learning (R)

Foundation Models & LLMs (Python)

Real Analysis I–II (Honors)

RESEARCH EXPERIENCE

Inequality of Opportunity: English-Medium Instruction in India 🇮🇳 🌐 🌍 Spring 2025–Fall 2026

Senior Thesis Paper (with feedback from Prof. Laura Schechter)

- Engineered original district data system (1961–2024, mapped to 2001 borders) incorporating 52 Census tables; 14 NSS, HCES, & PLFS surveys; 11 DISE censuses, 7 linguistic datasets, 4 CPI time series, 2 Economic Censuses.
- Built administrative histories for 593 districts using Census & LGD records, geographic evidence, and official state gazettes, reconciling districts over renames, mergers, splits, and typos. Harmonized 9 consumption surveys with varying questionnaires and recall periods by defining comparable MPCE measures, tested against published values.
- Reconstructed lost schooling data for 7,211 district-year-medium of instruction IDs by processing 22 archived DISE PDFs. Recovered lost lexicostatistical data to test robustness of linguistic distance proxies like Glottolog genealogy.
- Implemented state-FE and survey-weighted ANCOVA, probit, and IV models with extensive sensitivity analysis. Tested spatial autocorrelation via permutation Moran's I . Evaluated weak IV inference using Montiel Olea–Pflueger effective F and Anderson–Rubin confidence sets. Controlled family-wise error rate using Holm–Bonferroni method.
- Built reproducible `{targets}` pipeline in R with `renv`-locked dependencies, SHA-256 checksums for raw file manifests, automated tests, and end-to-end generation of publication-quality tables, maps, and manuscript.

Language Models as Measurement Instruments: Risks for Estimation

Summer 2026–Present

Independent Research Project (with feedback from Prof. Harold D. Chiang)

- Formulated LM-generated measures as noisy proxies for central banks' monetary policy stance. Decomposed noise into non-classical error and training leakage. Designed experiments isolating their effects on macro prediction.
- Built reproducible Python codebase for World Central Banks corpus (380,200 sentences, 3,453 docs, 1996–2024): ingestion, measurement, aggregation, FE regressions with IMF/World Bank data, and automated tables/figures.
- Evaluated non-classical measurement error using 25,000 human-labeled sentences and entity masking, semantic perturbations, validation sample-size simulations, regressions predicting errors, and sensitivity to aggregation rules.
- Currently fine-tuning RoBERTa model while using a TF-IDF/logit baseline to benchmark and debug the pipeline. Will compare models by predictive performance, potential propagation of errors into macroeconomic estimation.

Fast Inference for Large- N Panel Quantile Regression & Common Shocks

Spring 2026–Present

Research Project (with guidance from Prof. Harold D. Chiang)

- Defined stochastic subgradient estimator for fixed-effects panel quantile regression under common shocks. Demonstrated $\sqrt{T}$ -normality under full-period stochastic increments. Outlined Polyak–Ruppert averaging, online updating, and mini-batch rate conditions for $o_p(T^{-1/2})$ score residuals and stochastic approximation errors.
- Eliminated first-order effect of FE scores and N -dimensional incidental parameters via orthogonalized slope moment, yielding the Schur-complement Jacobian Γ and identifying the common-shock variance component Σ .
- Developed proof strategy for first-order equivalence between one-step corrected SGD estimator and exact FEQR: asymptotically negligible KKT residuals imply $\sqrt{T}$ -normality and valid $\Gamma^{-1}\Sigma\Gamma^{-1}$ sandwich inference.

Empirical Implications of Stochastic Hysteresis on Optimal Policy

Fall 2025

Independent Research Project (with guidance from Prof. Job Boerma)

- Derived observable proxies for Boerma et al. (2023)'s Malliavin/Itô calculus given CRRA utilities, CRS, etc. to distinguish true hysteresis from persistence or long memory in macro, finance, and climate policy.
- Developed framework to falsify large class of Markov models (e.g., DSGE, VAR) using non-Markov time series. Assessed computational and identification tradeoffs of Kalman filtering, particle filtering, and Bayesian MCMC.

Dynamic Discrete Choice in Fast Food: Loyalty, Technology, & Inequality Summer–Fall 2025

Independent Research Project (with feedback from Prof. Amanda Smith)

- Formulated and numerically optimized a structural model of discounts, channel choice (in-store vs. app vs. outside option), and loyalty adoption using exponential cone optimization (Julia/JuMP, MOSEK) with MNL demand.
- Built a reproducible simulation procedure (e.g., $T = 20$) with evolving states, feasibility/termination checks, & numerical stability via stable softmax. Outlined counterfactuals for COVID, input-cost inflation, inequality shocks.
- Ensured convexity despite MNL demand & concave technology investment via log-sum-exp & RSOC epigraphs. Avoided bilinearity by updating loyalty shares via lagged states. Added income segmentation via MILP Big-M.

WORK EXPERIENCE

Economic Consultant and Data Analyst: Capital and Firm Ownership in Retirement 2024

Maui Mastermind Business Coaching

Remote

- Wrangled the complex survey design and multiple imputation of the Federal Reserve's *Survey of Consumer Finances* to harmonize its microdata and study the role of business ownership in post-retirement wealth persistence.
- Estimated age-grouped wealth distributions via survey-weighted quantiles (P50-P99.99), with replicate-weight SEs combined under multiple imputation (5 implicates, 999 replicate weights), in R.
- Illustrated findings in client-readable tables/figures. Reported results alongside concise discussions of how SCF measurement, sampling, and data gaps affect interpretation, balancing technical precision with accessibility.

Restaurant and Inventory Manager 2022–2023

Sookie's Veggie Burgers

Madison, WI

- Managed a 10-person team in daily operations. Implemented data-driven inventory tracking to cut waste and holding times. Strengthened communication/teamwork under tight operational constraints. Designed menu items.

Tutor of College-Level Economics, Calculus, Chemistry, and US History 2022–2023

Self-Employed

Remote

- Improved student performance using self-made problem sets and exams, clear intuition, and engaging exposition.

EXTRACURRICULAR EXPERIENCE

Presenter, PREDOC.org Research Conference Summer 2026

Booth School of Business, The University of Chicago

Presenter, Undergraduate Research Symposium Spring 2025

University of Wisconsin-Madison

Policies for Labor Recovery: Economic Development Case Competition (1st Place) Fall 2024

Sole Author of Written Report & Collaborator on Presentation (with feedback from Prof. Simeon Alder)

- Analyzed local general equilibrium impacts of a coal-plant closure using demand spillovers, labor reallocation, BLS and WI DWD projections, and direct/indirect/induced labor/fiscal losses (\$251M output, \$2.8M taxes, 308 jobs).
- Designed demand-driven training addressing job-specific human capital, school funding, housing supply, & retention via supportive services, post-training mentorship. Evaluated funding, from WIOA to WI Fast Forward & SCSEP.
- Presented strategies and potential partnerships for improving business dynamism and climate (licensing, zoning, knowledge diffusion) given local demand gaps (e.g., 50% retail furniture leakage) to Alliant Energy, MadREP.

AWARDS & HONORS

Veldor Kopitzke Scholarship in Economics Fall 2025

Department of Economics, University of Wisconsin-Madison

Dean's List Fall 2023, Spring 2025, Spring 2026

University of Wisconsin-Madison

Leadership Recognition: President's Volunteer Service Award Winner (Field Museum),
Martin Luther King Jr. Outstanding Young Person Award (Urban League of Greater Madison),
Certificate of Distinguished Achievement for Outstanding Citizenship (American Legion)

TECHNICAL SKILLS

Proficient: R (incl. spdep), Python (incl. sklearn, transformers), Stata, Julia, MATLAB, SQL, Git, L^AT_EX

Familiar: C++, Fortran, JavaScript, HTML/CSS, Mathematica